3. Find $X$ and $Y$ if $X+Y=\begin{pmatrix}7 & 0\\ 3 & 5\end{pmatrix}$ and $X-Y=\begin{pmatrix}3 & 0\\ 0 & 4\end{pmatrix}$

4. If $A=\begin{pmatrix}0 & 4 & 9\\ 8 & 3 & 7\end{pmatrix}$, $B=\begin{pmatrix}7 & 3 & 8\\ 1 & 4 & 9\end{pmatrix}$ find the value of (i) $B-5A$ (ii) $3A-9B$

5. Find the values of $x,\ y,\ z$ if (i) $\begin{pmatrix}x-3 & 3x-z\\ x+y+7 & x+y+z\end{pmatrix}=\begin{pmatrix}1 & 0\\ 1 & 6\end{pmatrix}$

   (ii) $(x \quad y-z \quad z+3)+(y \quad 4 \quad 3)=(4 \quad 8 \quad 16)$

6. Find $x$ and $y$ if $x\begin{pmatrix}4\\ -3\end{pmatrix}+y\begin{pmatrix}-2\\ 3\end{pmatrix}=\begin{pmatrix}4\\ 6\end{pmatrix}$

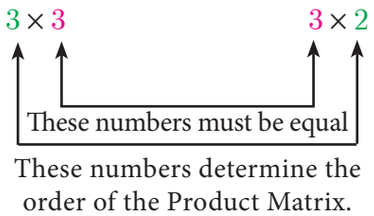
7. Find the non-zero values of $x$ satisfying the matrix equation

$$x\begin{pmatrix}2x & 2\\ 3 & x\end{pmatrix}+2\begin{pmatrix}8 & 5x\\ 4 & 4x\end{pmatrix}=2\begin{pmatrix}x^2+8 & 24\\ 10 & 6x\end{pmatrix}$$

8. Solve for $x,\ y$ : $\begin{pmatrix}x^2\\ y^2\end{pmatrix}+2\begin{pmatrix}-2x\\ -y\end{pmatrix}=\begin{pmatrix}5\\ 8\end{pmatrix}$

## Multiplication of Matrices

To multiply two matrices, the number of columns in the first matrix must be equal to the number of rows in the second matrix. Consider the multiplications of 3×3 and 3×2 matrices.



These numbers determine the order of the Product Matrix.

(Order of left hand matrix) × (order of right hand matrix) → (order of product matrix).
$(3\times3)$ $(3\times2)$ → $(3\times2)$

Matrices are multiplied by multiplying the elements in a row of the first matrix by the elements in a column of the second matrix, and adding the results.

For example, product of matrices $\begin{pmatrix}a & b\\ c & d\\ e & f\end{pmatrix}\times\begin{pmatrix}g & h & i\\ k & l & m\end{pmatrix}=\begin{pmatrix}ag+bk & ah+bl & ai+bm\\ cg+dk & ch+dl & ci+dm\\ eg+fk & eh+fl & ei+fm\end{pmatrix}$

The product $AB$ can be found if the number of columns of matrix $A$ is equal to the number of rows of matrix $B$. If the order of matrix $A$ is $m\times n$ and $B$ is $n\times p$ then the order of $AB$ is $m\times p$.

## Properties of Multiplication of Matrix

### (a) Matrix multiplication is not commutative in general

If $A$ is of order $m\times n$ and $B$ of the order $n\times p$ then $AB$ is defined but $BA$ is not defined. Even if $AB$ and $BA$ are both defined, it is not necessary that they are equal. In general $AB\neq BA$.



3 Algebra.indd 149 16-12-2020 01:07:28 PM